

UBT311: MOLECULAR BIOLOGY, FOOD SCIENCE AND IMMUNOLOGY LAB

L T P Cr

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Course objective: The course aims to make students understand and comprehend the basic concepts of molecular biology, food science and immunology and get hands-on experience with the basic techniques for determining food composition.

Detail contents:

Molecular Biology:

Bacterial transformation, Isolation of plasmid DNA, Genomic DNA and total RNA isolation from bacterial cells, UV spectrophotometric analysis of DNA samples, Restriction digestion and its analysis by Agarose gel electrophoresis / Native PAGE, Amplification of target DNA by PCR, Gene induction

Food Science:

Determining the RDA's and planning diet, Estimation of moisture content of foods, Extraction and estimation of fat content of foods, Determining protein content in food through Kjeldhal's method, Determine the quality of various foods (dairy, fruit and vegetable, bakery products), Sensory evaluation of foods

Immunology:

Performance of blood typing, Performance of latex agglutination, Quantification of antigen concentration using radial immunodiffusion method, Study of the reaction pattern of an antigen with a set of antibodies by Ouchterlony Double Diffusion method, Quantification of antigen concentration by Sandwich ELISA method, Isolation of peripheral blood mononuclear cells (PBMCs) from EDTA blood

Course Learning Outcomes (CLO)

The students will be able to:

1. transform bacterial cells with plasmid DNA, isolate plasmid, genomic DNA and total RNA.
2. perform facile techniques of molecular biology.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

3. determine food quality by food analysis.
4. apply the immunological concept and techniques for research purposes and industrial uses.

Textbooks

1. Rangana, S. Handbook of Analysis and Quality Control For Fruit and Vegetable Products.
2. Primrose, S.B. and Twyman, R.M., Principles of Gene Manipulation and Genomics, Blackwell Publishing (2006).
3. Punt J, Stranford S, Jones P, Owen J. Kuby- Immunology W.H. Freeman & Company (2019) VIIIth edition

Reference Books

1. Murphy K., and Weaver C. Janeway Immunobiology Garland Exclusive (2016) IXth edition

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	Sessionals (May include assignments / quizzes / performance)	100

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